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A paleomagnetic study of the Lower Cretaceous Cupido Limestone, northeast Mexico: Evidence for local rotation within the Sierra Madre Oriental

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ABSTRACT

We collected 401 oriented samples from the Lower Cretaceous Cupido Limestone in northeast Mexico at an average stratigraphic spacing of 2 m. Analysis of the remanent magnetization reveals an unstable component that is removed upon heating to 300 °C and a component that remains stable to 550 °C. Thermal demagnetization and isothermal remanent magnetization (IRM) studies indicate that the carrier of this stable remanence is magnetite. The magnetite is believed to be detrital and therefore responsible for a remanence associated with the time of deposition.

With the exception of a small number of spurious samples, all samples are normally magnetized. This is in contrast to studies elsewhere in which as many as three reversed intervals have been identified during the time of Cupido deposition.

The pole position obtained, 31.0°N, 169.8°W (α_{95} = 3.1°, k = 6.1, N = 401), lies some 35° to 40° south of other post-Jurassic Mexico pole positions, suggesting that the sampling site has rotated counterclockwise by a similar amount relative to stable Mexico. The sampling site is located in a northeast-trending portion of the Sierra Madre Oriental that offsets the main southeast-trending ranges by 50 km. The northeast trend is interpreted as the result of a "wrapping" of the folds around the southeast corner of the old Coahuila Peninsula, which acted as a barrier to Laramide décollement deformation. The counterclockwise rotation suggested by the paleomagnetic results therefore is thought to be due to local rotation of the folds and not related to major plate motions between North America and Mexico.

INTRODUCTION

Mexico's past relationship with "stable" North America is one of considerable controversy (Guerrero and Helsley, 1974; Nairn, 1976, 1978; Urrutia-Fucugauchi and Pal, 1977; Pal and Urrutia-Fucugauchi, 1978; Gose and others, 1980; Walper, 1980; and Gose and others, 1982). Obtaining paleomagnetic pole positions for Mexico that cover the geologic time scale is difficult because of insufficient exposures, unclear stratigraphic correlations, and local tectonism. The first problem has hindered studies in central and northwest Mexico, whereas the last two problems apply particularly to the Mesozoic strata of northeast Mexico. The Cretaceous limestones in the Sierra Madre Oriental in the vicinity of Saltillo are well exposed and stratigraphically well defined, thus making them well suited for this study.

The intent of this study is twofold: (1) to determine an Early Cretaceous pole position for northeast Mexico, and (2) to examine the Cretaceous long normal (Mercanton) magnetic interval in order to identify reversal events that have been reported in this part of the stratigraphic column elsewhere (Channell and others, 1979; Lowrie and others, 1980) and also in oceanic magnetic anomalies (Larson and Hilde, 1975).

The work presented here is based on a master's thesis (Kleist, 1980) and is a more detailed treatment of material previously published in abstract form (Kleist and others, 1980).

GEOLOGIC SETTING

The Sierra Madre Oriental Mountains of northeastern Mexico are the products of an early Tertiary (Laramide) orogenic event and extend from the United States–Mexican border to the northern part of Central America. This orogenic belt is composed principally of deformed Mesozoic sedimentary rocks, and the Cretaceous limestones form most of the topographic high ground. The main structural trend of the Sierra Madre Oriental is northwest-southeast, but a significant bend in this trend near Monterrey produces an almost east-west orientation (Fig. 1). This paper examines the Lower Cretaceous Cupido Limestone in outcrops within this east-west trending part of the orogen. Stratigraphically, the Cupido Limestone conformably overlies the Late Jurassic–Neocomian La Casita Formation that is, in part, the facies equivalent of the Taraises Formation (Wilson and Pialli, 1978). The Cupido in turn is conformably overlain by the upper Aptian La Pena Formation. These bounding formations constrain the time of Cupido deposition from early Barremian to early Aptian. The Cupido itself contains very few biostratigraphic markers and there is no lithologic evidence for unconformable contacts or long hiatuses (J. L. Wilson, 1980, personal commun.).

The Cupido Limestone in the study area of Arteaga Canyon consists primarily of resistant, thick-bedded (50–100 cm) dark gray lime mudstones and wackestones with very thin shale interbeds. In the regional paleoenvironmental studies of Conklin and Moore (1978) and of Wilson and Pialli (1978), the Arteaga section was interpreted to lie entirely within a lagoonal facies. The carbonate platform that was the site of Cupido deposition developed on the

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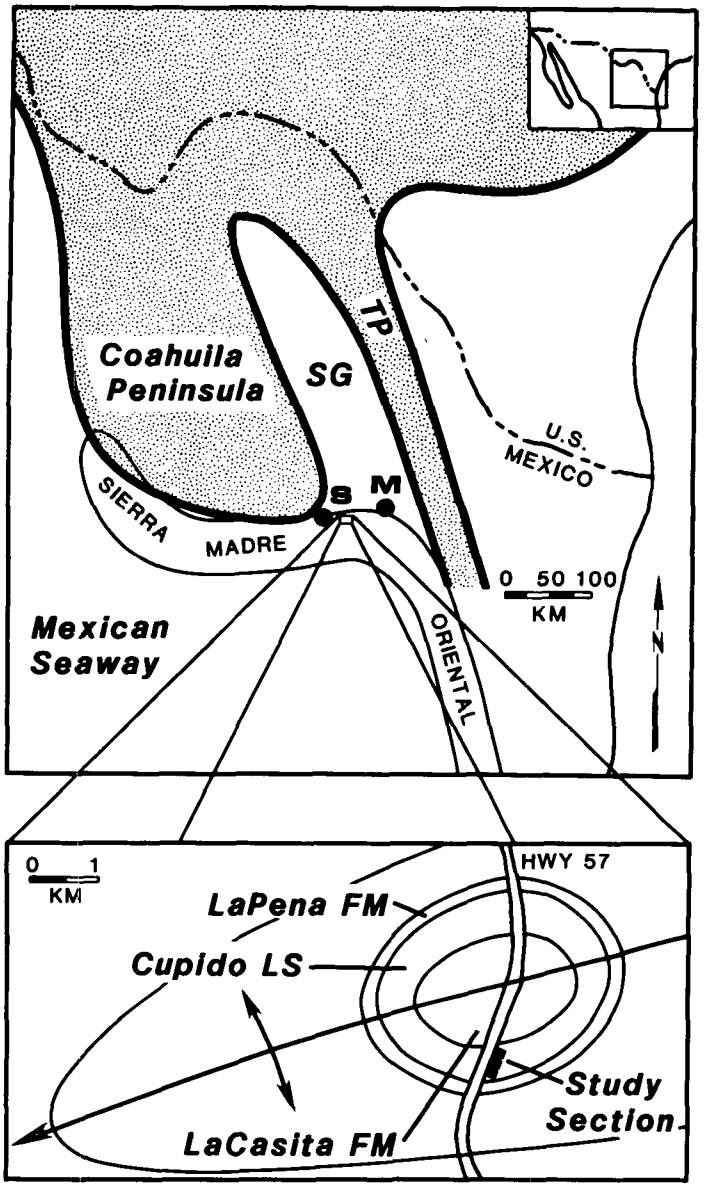


Figure 1. Late Jurassic paleogeography of northeast Mexico relative to present-day geography and the Sierra Madre Oriental, with the location of the Cupido sampling site (inset). Geologic map from de Cserna (1956). SG = Sabinas Gulf, TP = Tamaulipas Peninsula, S = Saltillo, M = Monterrey, FM = Formation, LS = Limestone, HWY = Highway.

margins of the Coahuila Peninsula, a positive feature that existed through much of the Cretaceous (Fig. 1). During the Eocene, northeastern Mexico was subjected to deformation that resulted in the Sierra Madre Oriental orogenic belt. The trend of this orogen is believed to reflect the previously existing paleogeographic elements, including the Coahuila Peninsula (Humphrey, 1949).

FIELD AND LABORATORY METHODS

The Cupido Limestone was sampled approximately 20 km southeast of Saltillo, Coahuila, in Arteaga Canyon (25°22′N, 100°49′W). The Arteaga Canyon section is exposed in the south flank of the Anticlinal del Chorro, part of the Arteaga Anticlinorium (Fig. 1). Excellent access is provided by Highway 57, which transects the anticline in a north-south direction. The Anticlinal del Chorro is a westward-plunging anticline with an axial trend of N75° E at the sampling site. The amount of plunge in the Anticlinal del Chorro in the area of this study is less than 5° (calculated from the change of elevation along the fold axis on 1:50,000 aerial photographs and topographic maps of the area). The effect of this amount of plunge on the geometric reconstructions therefore is negligible. A 750-m–thick section is exposed in road-cuts and weathered slopes along the highway. We collected 401 oriented samples at an average spacing of 2 m. Although it is unlikely that the section studied formed by constant and continuous deposition, field evidence indicates no major breaks in deposition. Thus, as Cupido deposition spans about 9 to 10 m.y. (Barremian to late Aptian), the sampling plan was designed to monitor the behavior of the geomagnetic field at approximately 25,000-yr intervals. Such sampling is believed to be sufficient to easily resolve the reported intervals of reverse polarity of 0.5 m.y. or greater.

TABLE 1. SUMMARY STATISTICS OF MAGNETIC RESULTS

Treatment	Number of samples	D_m (degrees)	I_m (degrees)	k	α_{95} (degrees)
(°C)					
NRM					
w/structural corr.	422*	154.8	66.3	9.6	2.3
w/o structural corr.	422*	348.2	43.7	10.1	2.3
300 all samples†	401	292.6	49.3	6.1	3.1
220 *pilot group†	90	285.8	51.3	4.2	8.3
300 pilot group†	90	289.9	44.2	4.5	7.9
350 pilot group†	90	294.4	52.3	5.2	7.2
400 pilot group†	90	294.0	52.3	5.2	7.2
450 pilot group†	90	286.8	64.9	2.7	11.8
500 pilot group†	45	265.9	56.4	2.4	17.6
550 pilot group†	45	208.8	60.5	1.8	23.4
600 pilot group†	45	84.3	-11.1	1.3	41.1
650 pilot group†	9	132.2	9.5	2.0	51.1
700 pilot group†	9	306.4	64.3	1.8	57.4

*Includes 21 replicate samples (2 samples from 1 core).
†With structural correction.

The magnetization of the samples was measured at the University of Texas Paleomagnetic Laboratory using a two-component, cryogenic magnetometer housed in a shielded room. The optimum thermal-treatment temperature was established by use of 90 pilot samples (20% of the total). Groups of samples were selected at 20-m intervals, and the following factors were considered in this selection process: (1) The lithological character of the Cupido Limestone is uniform. No major lithologic variations occur in this section. (2) Variations in natural remanent magnetization (NRM) intensities show no systematic change with stratigraphic position.

The pilot samples were stepwise thermally demagnetized, first at 220 °C and then at 50 °C intervals from 300 to 700 °C. The sample directions for representative steps to 600 °C are shown in Figure 2 and the Fisher statistics for each step are given in Table 1. The behavior of the samples during demagnetization shows two major components of remanence: (1) an approximately present-day field component that dominates the NRM but that is effectively removed in the first heating step (Figs. 3a, 3b), and (2) a component that remains stable to approximately 500 °C (Table 1). There is a general loss in directional stability at and above 600 °C, which suggests that magnetite is the carrier of this second component. On the basis of the intensity and directional stability, 300 °C was chosen as the optimum level of demagnetization. The median inten-

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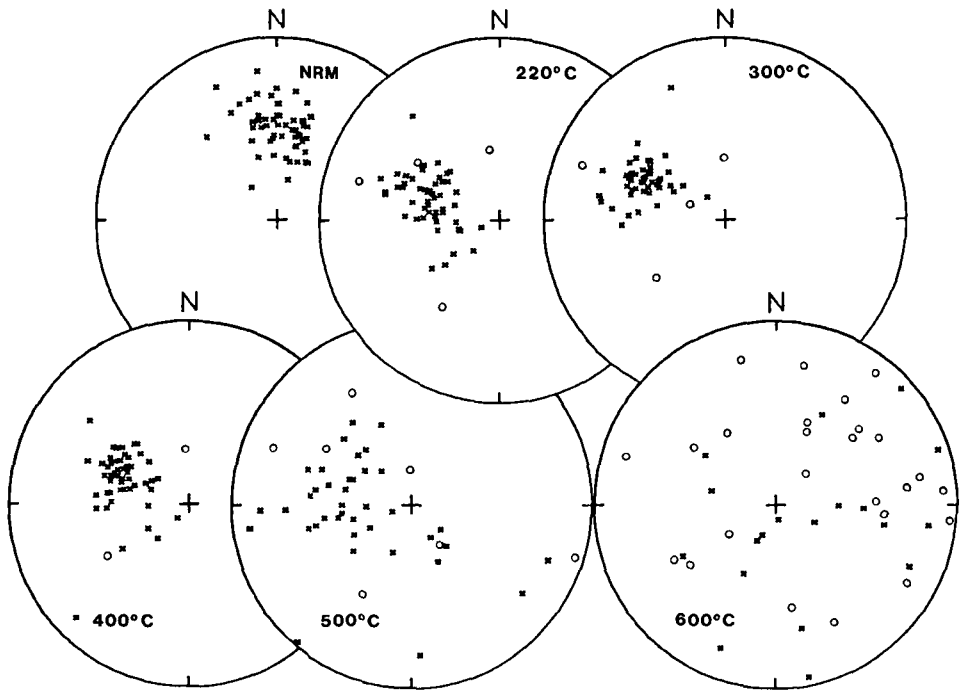


Figure 2. Example stereographic plots of magnetization directions for pilot samples at NRM (without structural correction) and after increasing levels of thermal demagnetization (with structural correction). Crosses and circles represent positive and negative inclinations, respectively.

sity at this step is about 2.5×10^{-7} emu cm^{-3}, well above the noise level of the magnetometer (namely, 2×10^{-8} emu cm^{-3}).

Several Cupido samples were exposed to stepwise-increasing magnetic fields and the resulting isothermal remanent magnetization (IRM) was measured. In fields of 200 to 600 Oe, an IRM that levels off between 1,000 and 2,000 Oe (Fig. 3c) is rapidly acquired. Above 2,000 Oe, the IRM begins to increase once more. The plateau in the IRM acquisition curve is indicative of the presence of magnetite in the samples that saturate at low fields. The increase beyond 2,000 Oe may represent the contribution from hematite and/or goethite, both of which have high coercivities. Goethite was identified by X-ray diffraction, but hematite was not. The low-temperature component of the remanence is probably due to this goethite, which forms as a result of surface weathering. The IRM acquired in a 3,000-Oe field, the maximum field attainable in the laboratory used, was removed by stepwise thermal demagnetization (Fig. 3d). The intensity and directional stability again decrease markedly between 550 °C and 600 °C, indicating that magnetite dominates this IRM.

MAGNETIC STRATIGRAPHY

Of the 401 samples measured at 300 °C, only 4 give reliable magnetizations with reversed polarity. No two of these samples are stratigraphically successive and may be the result of inadvertent inversion in the field or laboratory. Certainly there is no interval over which several reverse magnetizations are observed. This is in marked contrast to the studies of Channell and others (1979) and of Lowrie and others (1980), in which three reversals were identified, two in the Barremian and one in the early Aptian. These reversals appear to be approximately 0.4 to 0.6 m.y. in duration. The absence of reversals in our study may be due to (1) periods of nondeposition corresponding to the times of reversed field, (2) remagnetization of the section during a period of normal polarity, and (3) inadequate sample spacing that has resulted in the reversed intervals being missed.

As three reversals are reported to be present in the time interval over which the section was formed, it is extremely unlikely that there would be major periods of nondeposition that so exactly coincide with times of reversed field. In addition, there is no evidence in the section to demonstrate any major breaks in deposition. On the basis of the thermal demagnetization and IRM studies, magnetite is believed to be the carrier of stable remanence. Studies of depositional environments similar to that of the Cupido (Cloud, 1962) indicate that, although reducing conditions are present at shallow depths, these conditions are not sufficient for the formation of authigenic magnetite. The magnetite present therefore is more probably detrital in origin and consequently the remanence is that associated with the geomagnetic field at the time of deposition. This suggests that the section has not undergone extensive remagnetization. Resetting the magnetite magnetization by other means also appears to be unlikely. Finally, the sampling plan is based not upon the assumption of continuous deposition but rather on the hypothesis that any time lost due to nondeposition is more or less uniformly distributed throughout the bedding planes. As the reversed intervals reported are much longer than 25,000 yr, it is difficult to envisage how our sampling plan would miss such significant reversals.

In the absence of evidence to the contrary, we therefore interpret our results as indicating that the entire Cupido section was deposited in a period of constant normal polarity.

POLE POSITION

Although the age of the stable magnetization is uncertain, the results of the fold test shown in Figure 4 clearly demonstrate that it predates the folding of the strata. Furthermore, the presence of magnetite as the dominant carrier of stable remanence suggests that the magnetization is primary and consequently approximately syn-

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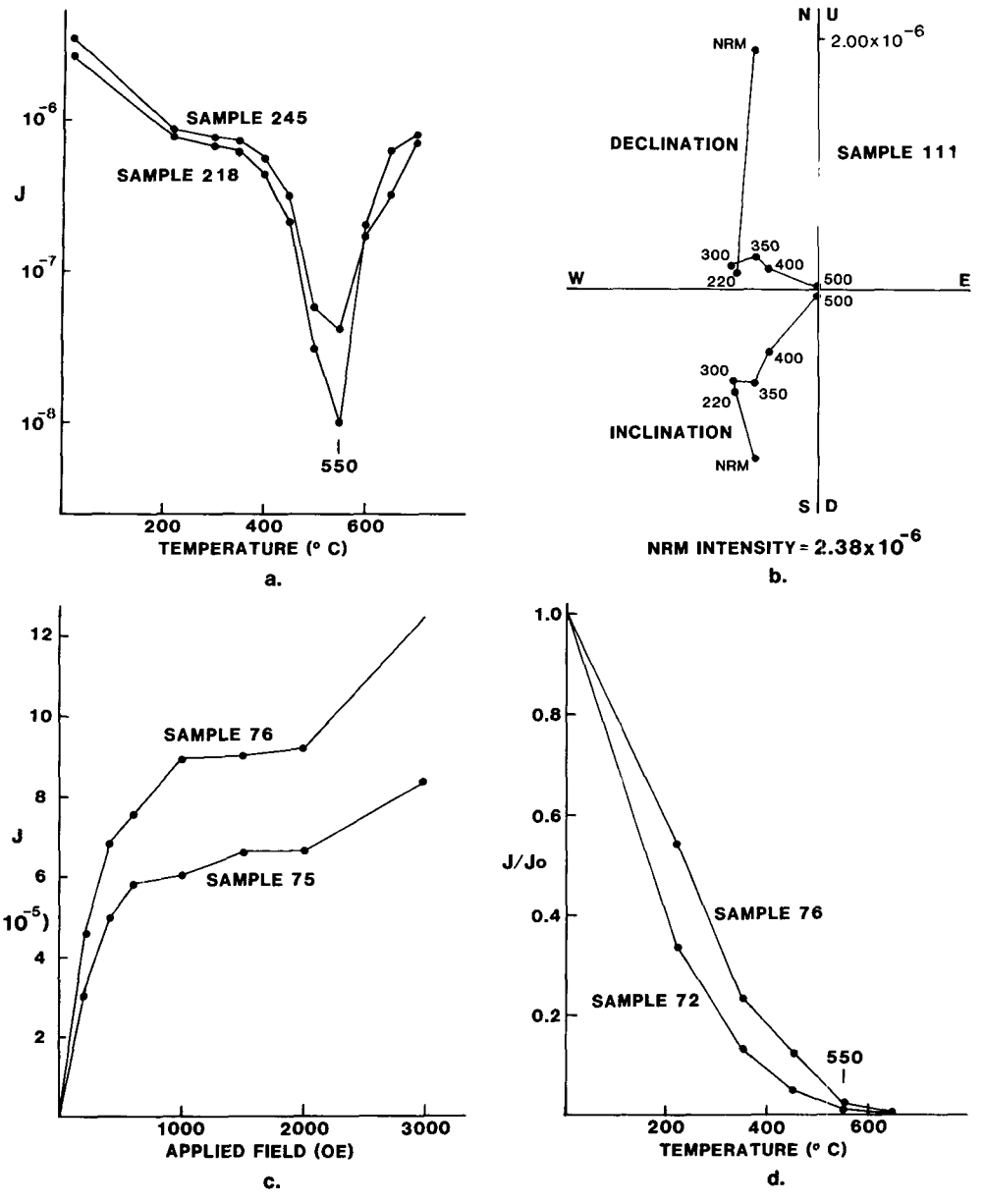


Figure 3. a. Examples of changes in remanence intensities during thermal demagnetization of NRM. J = intensity (emu/cm^3). b. Example vector plot showing remanence direction and intensity changes during thermal demagnetization. Intensities are normalized to NRM intensity. c. Example of IRM acquisition curves. d. IRM thermal demagnetization curves. J/J_0 is the ratio of remaining intensity to intensity after exposure to 3,000-Oe field.

depositional. The paleomagnetic pole obtained (31.0°N, 190.2°E, α_{95} = 3.1°, k = 6.1, N = 401), therefore, is believed to give an early Aptian pole position for the sample site. This pole is plotted in Figure 5 together with an apparent polar wander path for Mexico (Urrutia-Fucugauchi, 1979). The Cupido pole lies some 35° south-southeast of the Mexican poles of similar age, suggesting a counterclockwise rotation of the sampling site by a similar amount. Also shown in Figure 5 is the result of rotating the Cupido pole clockwise by 35°. This rotation places the "restored" pole more or less coincident with the established polar wander curve for Mexico.

The rotation is attributed to the Eocene Laramide Orogeny because this was the only major diastrophic event after Cupido deposition. Laramide deformation in northeast Mexico is described by de Cserna (1956) as a sliding of the sedimentary cover to the northeast along décollement horizons. Crustal shortening during this sliding has been estimated to be about 50% in the Monterrey area (Baker, 1941). Examination of the trend of the Sierra Madre Oriental in the Saltillo-Monterrey area (Fig. 1) shows that the anticlines have an arcuate shape, ranging from a northwest trend south of Monterrey to a southwesterly trend near Arteaga Canyon. The overall trend of the Sierras is believed to reflect the margins of the old Coahuila Peninsula (Fig. 1) and other positive features during the Cretaceous. King (1942) stated that the folds were "wrapped around the opposing buttress of the Coahuila Peninsula." In essence, this early observation aptly describes the proposed mechanism for rotation of the Arteaga sampling site, which was located on the southeast corner of the Coahuila Peninsula. At the onset of Laramide deformation, the initial structural trend paralleled the

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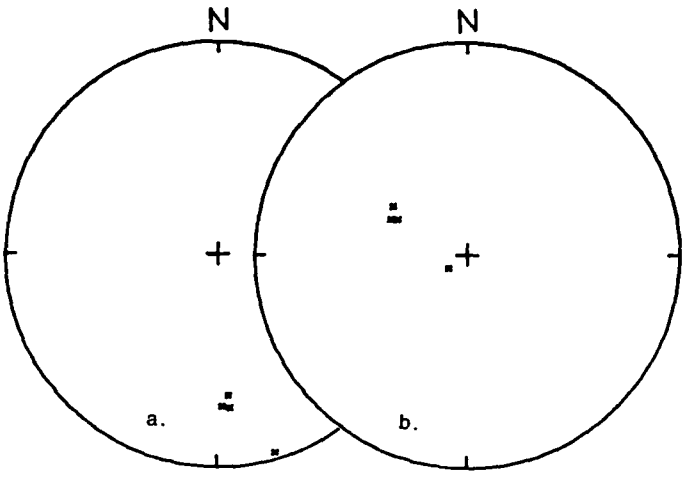


Figure 4. Fold test stereonets for Cupido samples from the north flank of the El Chorro anticline. a. Magnetization directions without structural correction. b. Magnetization directions with structural correction. Samples thermally demagnetized at 300 °C.

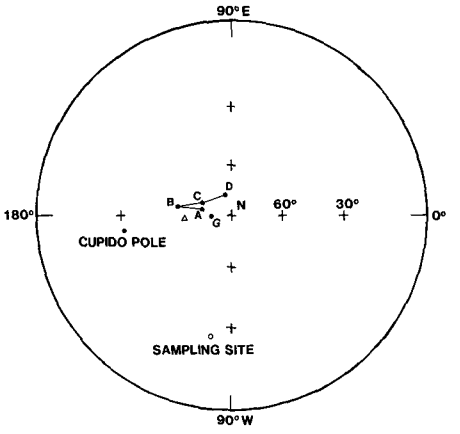


Figure 5. Polar stereographic projection showing the Cupido pole position and a portion of Mexico's apparent polar wander path (from Urrutia-Fucugauchi, 1979). Pole ages are: A, Tertiary; B, Late Cretaceous (Difunta); C, Early Cretaceous; D, Late Jurassic; G, Aptian-Albian pole of Gose and others (1982). Open triangle denotes "restored" Cupido pole.

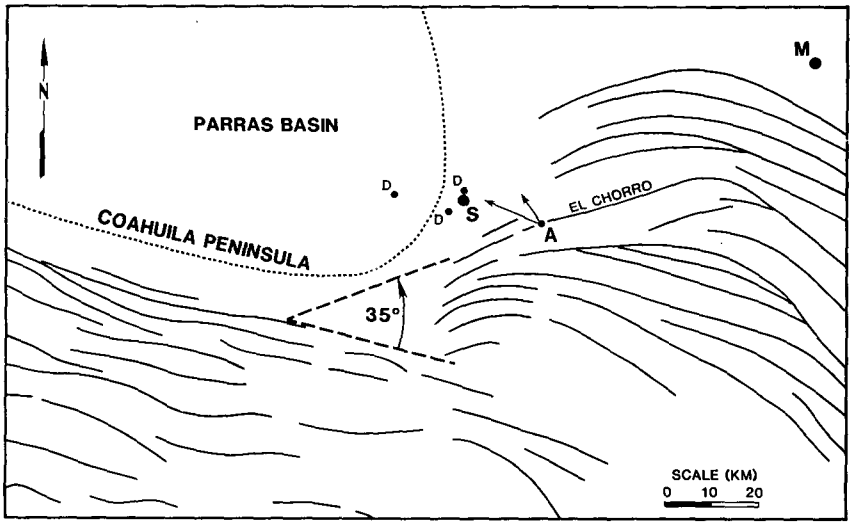


Figure 6. Anticlinal axes of the Sierra Madre Oriental near the southwest corner of the old Coahuila Peninsula (from de Cserna, 1956). A = Arteaga Canyon sampling site, S = Saltillo, M = Monterrey, D = Nairn's (1976) Difunta sampling sites. Long arrow at Arteaga Canyon represents mean declination of Cupido samples; short arrow represents estimated prerotation declination.

southern margin of the Coahuila Peninsula, crossing the opening to the Sabinas Basin and turning more southeasterly along the shallower basement of the old Tamaulipas Peninsula. With continued compression, the deeper basement of the Sabinas Basin allowed more northward sliding of the sedimentary cover, resulting in a rotation of the folds around the southeast corner of the old Coahuila Peninsula (Fig. 6). The angular change between the fold trend south of the peninsula and that at Arteaga Canyon is approximately 35°, the amount of rotation indicated by the Cupido pole position.

Comparison of the Cupido pole position with that from the Late Cretaceous Difunta Group near Saltillo (Nairn, 1976) gives some indication of the areal extent of rotation. The Difunta pole position (57.9°N, 169.3°E, A_{95} = 8.4°) suggests a possible rotation that is small compared to that of the Cupido site. The Difunta sampling sites, however, do not lie in the main fold belt (Fig. 6), but within the less intensely folded Parras Basin strata that were deposited upon the Coahuila Peninsula. On the basis of this comparison, it is believed that the major rotation was restricted to the main fold belt.

DISCUSSION

The paleomagnetic evidence for local rotations associated with Laramide tectonism holds many implications for past and future studies. Nairn (1976) noted the possibility of local rotational movements as a problem in his data interpretation. Care must be taken in the interpretation of paleomagnetic data in Mexico

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because the effects of local rotations may in part mask pre-Laramide events that have been responsible for microplate (basement-involved) motions during the early history of the evolution of Mexico. Such pre-Laramide events include the early Mesozoic 130° counterclockwise rotation of a large part of northern Mexico reported by Gose and others (1982). In this instance, an Aptian-Albian pole position (Fig. 5) does not indicate any Laramide rotation.

There is much confusion about if, when, and where relative movement between Mexico and "stable" North America took place. Reconstructions by Gose and others (1980) and Urrutia-Fucugauchi (1981) suggested that the Texas lineament possibly represents the discontinuity, but they noted that this is far from certain. Walper (1980) was more specific in citing a system of megashears in Mexico as being the loci of movement during Laramide tectonism. Most of his left-lateral strike-slip displacement (150 km) was attributed to the "Torréon-Monterrey megashear" that passes along the southern margin of the Coahuila Peninsula and was used to explain the "deflection" of the fold belt there. However, the mechanism outlined in the present study describes the sinuous trend of the fold belt as reflecting the margins of older paleogeographic barriers (for example, Coahuila Peninsula) and, as such, better explains the continuous, as opposed to offset, nature of the Sierra Madre Oriental.

Whereas most studies suggested pre-Laramide or syn-Laramide motion between Mexico and North America, Urrutia-Fucugauchi (1981) found paleomagnetic evidence for relative rotation as recently as post-Oligocene. However, restoring the Cupido pole according to the angular offset in the fold belt implies that northern Mexico has not undergone any major rotations (greater than 10°) relative to North America since the Eocene.

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